

Question 1: Genetic Algorithm

Given:

- Objective Function:
[
 $f(x)=x^2$
]
- Initial Population:
[
 $\{2,4,6,8\}$
]

Step 1: Calculate Fitness Value

Fitness is calculated using:

$$\text{Fitness} = f(x) = x^2$$

For $x = 2$

$$f(2) = 2^2 = 4$$

For $x = 4$

$$f(4) = 4^2 = 16$$

For $x = 6$

$$f(6) = 6^2 = 36$$

For $x = 8$

$$f(8) = 8^2 = 64$$

Fitness Table

Individual (x)	Fitness (x ²)
2	4
4	16
6	36
8	64

Step 2: Selection

Choose the individuals having the highest fitness values.

- Individual = **8**, Fitness = **64**
- Individual = **6**, Fitness = **36**

Selected Parents: 8 and 6

Step 3: Convert to Binary

6 = 0110

8 = 1000

Step 4: Perform Crossover

Exchange the Least Significant Bit (LSB).

Before Crossover

Parent 1 = **0110**

Parent 2 = **1000**

Exchange the last bit.

Parent 1 : 0110

Parent 2 : 1000

↑
Exchange LSB

After Crossover

Offspring 1 = **0110 = 6**

Offspring 2 = **1000** = **8**

Since both parents have **0** as the last bit, the offspring remain unchanged.

Step 5: Mutation

Mutation randomly changes one bit in a chromosome.

Example:

Before Mutation

0110 (6)

After Mutation

0111 (7)

Mutation helps to:

- Increase population diversity.
- Prevent premature convergence.
- Escape local optimum solutions.
- Improve the chances of obtaining the global optimum.

Final Answer

Fitness Values

Individual Fitness

2	4
4	16
6	36
8	64

Selected Individuals: 6 and 8

Offspring after Crossover: 6 and 8

Role of Mutation: Mutation increases diversity, avoids local optimum solutions, and improves the search for the best solution.

Question 2: Perceptron Model

Given

Advertisement Exposure,

$$\begin{bmatrix} x_1=2 \end{bmatrix}$$

Income Level,

$$\begin{bmatrix} x_2=1 \end{bmatrix}$$

Weights,

$$\begin{bmatrix} w_1=0.5, \quad w_2=0.3 \end{bmatrix}$$

Bias,

$$\begin{bmatrix} b=-0.4 \end{bmatrix}$$

Step 1: Calculate Net Input

Formula:

$$\begin{bmatrix} \text{Net Input} = w_1x_1 + w_2x_2 + b \end{bmatrix}$$

Substitute the given values:

$$\begin{bmatrix} = (0.5 \times 2) + (0.3 \times 1) + (-0.4) \end{bmatrix}$$

$$\begin{bmatrix} = 1.0 + 0.3 - 0.4 \end{bmatrix}$$

$$\begin{bmatrix} = 0.9 \end{bmatrix}$$

]

Net Input = 0.9

Step 2: Apply Step Activation Function

Step Activation Function:

[
f(x)=
\begin{cases}
1, & x \geq 0 \\
0, & x < 0
\end{cases}
]

Since,

[
0.9 \geq 0
]

Therefore,

[
f(0.9)=1
]

Step 3: Decision

Output = 1

Therefore,

The customer will purchase the product.

Final Answer

Net Input

[
0.9
]

Activation Function Output

[
1
]

Prediction

Customer Purchases the Product.